Research question:

How transferable are fake-news detection signals across typologically different languages (English, Chinese, Japanese), and do **multilingual transformer** models generalize better than **linguistic feature-based models** in cross-language settings?

Specifically:

- How well do fake-news detection models trained in one language perform in another?
- Which linguistic features of fake news are universal vs language-specific?
- Do multilingual pretrained models outperform feature-based models in cross-lingual transfer?

Literature review:

Misinformation detection has been a popular subject of research, enabled by various machine learning, natural language processing, and now, more advanced deep learning techniques (Ali et al., 2025, 2025; Chu et al., 2021; Verma et al., 2021). Feature-based methods, such as those proposed in WELFake: Word Embedding Over Linguistic Features for Fake News Detection, combine word embeddings with stylistic and psycholinguistic indicators to classify misinformation efficiently. These methods highlight interpretable signals such as readability, exaggeration, and writing patterns. However, a majority of these techniques require large text datasets annotated by domain experts. Much existing scholarship and techniques of misinformation detection are built on English-language sources, given the abundance of available data, but similar research on other languages remains less available. Existing scholarships on cross-language fake-news detection have investigated the transferability of misinformation detection models between Chinese and English languages, utilizing multilingual transformer models or transfer learning of existing models trained on English-language sources. (Chu et al., 2021; Ozcelik et al., 2023; Yun et al., 2025). This thesis builds on these approaches by combining feature-based and multilingual transformer methods to examine transferability across English, Chinese, and Japanese, three languages with major structural and cultural differences.

Concept definition:

- **Misinformation:** used interchangeably with fake news/false information/etc. Broadly referring to all untrue news reports that contain false or misleading information.
- **Cross-lingual transferability:** the extent to which a model trained on one language performs well on data in another language without retraining

Research method & justification:

The study proposes two model families for misinformation detection across Chinese, English, and Japanese languages:

1. **Feature-based machine learning** (inspired by WELFake), including models such as logistic regression, SVM, and other well-performing models according to previous literature.
    a. Justification: interpretability, computationally efficient
2. **Multilingual transformer models,** such as multilingual BERT, will be fine-tuned for fake-news classification.
    a. Justification: more advanced methods, designed for cross-lingual transfer

The (tentative) linguistic features will follow the state-of-the-art categorization of:

- Readability index - word length, number of syllables, and sentence length.
- Psycho-linguistic - emotions, behaviors, persona, and mindset.
- Stylistic - style of a sentence.
- Quantity - sentence information, such as the number of words and the number of sentences.

Experiments will include

1. Within-language training & testing of ML models (baseline)
2. Cross-language zero-shot transfer (train in one language, test in another)
3. Fine-tuning transfer

Datasets:

English:

- WELFake dataset for fake news detection in text data
- Dataset used by WELFake, combining data from four different sources (BuzzFeed, Reuters, McIntire, and Kaggle), containing 72,134 news articles with 35,028 real and 37,106 fake news.

Chinese:

- [2403.09092] MCFEND: A Multi-source Benchmark Dataset for Chinese Fake News Detection
- contains 23,789 news pieces (though less balanced across the number of real vs. fake news)

Japanese:

1. Japanese Fake News Dataset | Taichi Murayama
    a. Fine-grained labeling capturing various aspects of fake news, containing 307 news stories and detailed annotations.
    b. **Issue: all annotations on the specific dimension of the fake information will need to be collapsed as simply "fake", losing nuance of the original data labels (e.g. misinformation vs. disinformation)**
2. GitHub - tanreinama/Japanese-Fakenews-Dataset: 日本語フェイクニュースデータセット
    a. Consists of news articles in Japanese and deep-fake articles generated by GPT-2 Japanese; data labeled as 0: original, 1: partially fake, and 2: completely fake
    b. **Issue: no guarantee of the authenticity of the 'original' news. The 'fake news' are also generated by AI, which may confound the real linguistic features of fake news. However, given the limited availability of Japanese-language datasets, this dataset, combined with the first dataset, can serve as a preliminary benchmark for the purpose of this experimentation.**

Additional Considerations and Limitations:

1. Dataset size imbalances
2. Potential difference in 'news' domains
3. Variation in labeling standards

However, ultimately, I believe this study is still valuable in its attempt to explore the transferability of models trained on readily available datasets to languages with less available data and to examine systematic differences in linguistic features across languages.